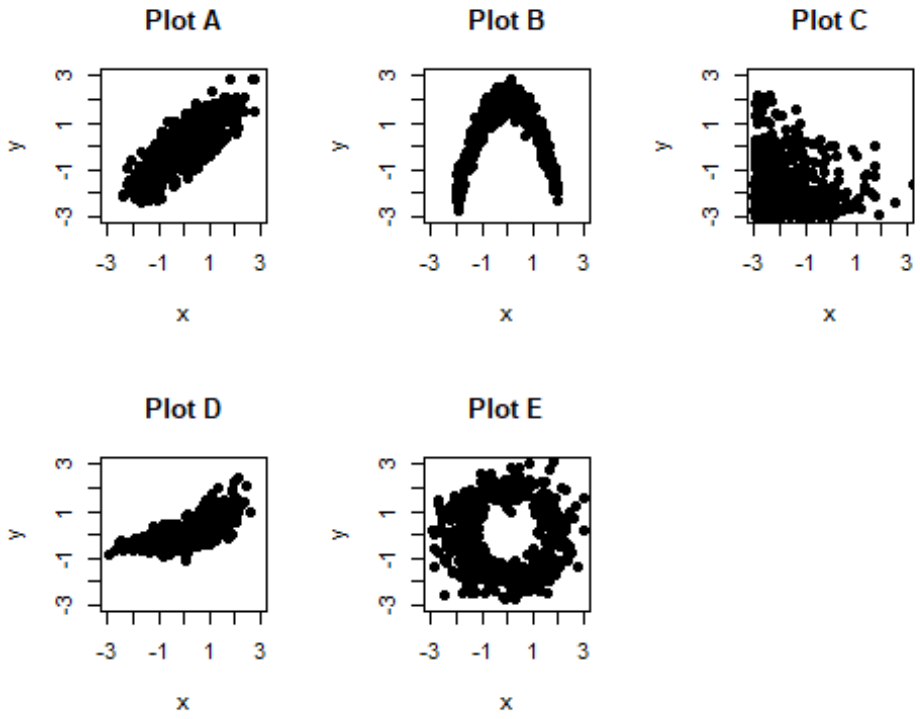
Review Questions for Lecture 8

You will want to use the R code in lecture07_code_updated.R to help you answer the questions that refer to the 5 plots below:



1. In plot A,
   a. Sketch (roughly) a histogram of the marginal distribution of y.
   b. Sketch (roughly) a histogram of the marginal distribution of x.
   c. Sketch (roughly) a histogram of the conditional distribution of y when $x \approx -1$.
   d. Sketch (roughly) a histogram of the conditional distribution of y when $x \approx 0$.

2. In plot B,
   a. Sketch (roughly) a histogram of the marginal distribution of y.
   b. Sketch (roughly) a histogram of the marginal distribution of x.
   c. Sketch (roughly) a histogram of the conditional distribution of y when $x \approx -1$.
   d. Sketch (roughly) a histogram of the conditional distribution of y when $x \approx 0$.
   e. Sketch (roughly) a histogram of the conditional distribution of x when $y \approx -1$.
   f. Sketch (roughly) a histogram of the conditional distribution of x when $y \approx 2$.

3. In plot C,
   a. Sketch (roughly) a histogram of the marginal distribution of y.
   b. Sketch (roughly) a histogram of the marginal distribution of x.
   c. Sketch (roughly) a histogram of the conditional distribution of x when $y \approx -3$.
   d. Sketch (roughly) a histogram of the conditional distribution of x when $y \approx -1$.

4. In plot D,
   a. Sketch (roughly) a histogram of the marginal distribution of y.
   b. Sketch (roughly) a histogram of the marginal distribution of x.
   c. Sketch (roughly) a histogram of the conditional distribution of y when $x \approx -1$.
   d. Sketch (roughly) a histogram of the conditional distribution of y when $x \approx 1$.

5. In plot E,
   a. Sketch (roughly) a histogram of the marginal distribution of y.
   b. Sketch (roughly) a histogram of the marginal distribution of x.
   c. Sketch (roughly) a histogram of the conditional distribution of y when $x \approx -2$.
   d. Sketch (roughly) a histogram of the conditional distribution of y when $x \approx 0$.

6-10. Estimate the mean and SD in plots A-E. Note that a quick and dirty way to estimate the standard deviation is as $range/4$. This is because most values will be between -2 and 2 in standard units if the data is reasonably bell-shaped.

11-15. Sketch the graph of averages (of y conditional on x) for plots A-E.

16-20. Sketch the regression lines for plots A-E.

21. Suppose that the scores on Midterm 1 have a mean of 140 and a standard deviation of 20, while the scores on the Final have a mean of 130 and a standard deviation of 30. Suppose also that the correlation between the midterm and final scores is 0.4 and that the scatter plot is football-shaped. Let x be the midterm score and y the final score.

   a. Determine the equation of the regression line that predicts y in terms of x.
   b. Determine the root mean squared error of predicting y based on x.
   c. Predict the score on the final for someone who scored 120 on Midterm 1.
   d. We would expect that roughly 95% of the scores on the final would be in what range for someone who scored 120 on Midterm 1?
   e. Determine the equation of the regression line that predicts x in terms of y.
   f. Is the equation in e) the same as the equation in a) after solving for x in terms of y?
   g. Does it make sense to use the equation in part a) to predict x based on y? Why or why not?

h. Suppose that the minimum score on Midterm 1 was 100. You use the equation in part a) to predict the score on the Final for someone who scored 80 on Midterm 1. What assumption are you making in making this prediction?
i. Do you think that the assumption you are making in part h) is reasonable? Why or why not?

22. Suppose that x and y have the same standard deviation. Under what conditions would the regression line for predicting x based on y have the same equation as the regression line for predicting y based on x after solving for x in terms of y?

23. This example will give some additional justification to the intuition of what happens to the correlation when you combine two datasets. Suppose that $SD(x_1) = SD(x_2), SD(y_1) = SD(y_2)$, and $corr(x_1, y_1) = corr(x_2, y_2)$, where the subscript indicates whether the variable comes from the first or second dataset.

a. Show that the slope of the regression line and the root mean squared error is the same in the two datasets.
b. Suppose further that $\overline{x_1} = 0$, $\overline{x_2} = 10$, and that the ranges of the *x* variable in the two datasets are -5 to 5 and 5 to 15, respectively. What is the relationship between $\overline{y_1}$ and $\overline{y_2}$ if the regression lines in the two datasets are the same?
c. Suppose now that the two datasets have the same regression line. Show that the regression line for the combined dataset is the same as the regression line for the individual datasets. (Hint: one approach would be an argument by contradiction, where you assume that the regression line is not the regression line in the individual datasets and then derive a contradiction. You will want to use the fact that the regression line minimizes the mean squared error.)
d. The result in part c) implies that the root mean squared error for the combined dataset is the same as the root mean squared error for the individual datasets. What will happen to the standard deviation of *y* in the combined dataset?
e. What does the result in d) imply about the correlation between *x* and *y* in the combined dataset? Justify your answer.